Capstone Dataset and Baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for Capstone Dataset and Baseline before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Model Cards and Human Review

Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Subgroup Performance

Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learning Curves

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Capstone Dataset and Baseline trace and record the first step you can explain without help.

Use W2: Define fields for Capstone data and **requirement**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Capstone Dataset and Baseline explanation against the supplied facts before discussing it.

Use the worked case for Capstone data and **requirement**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

A proposed 80-row dataset has 40 related pairs, and the team splits individual rows randomly then uses pair ID as a feature. Revise its plan.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Capstone Dataset and Baseline target to a trace and a justified conclusion.

A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and **baseline** procedure. When may final labels be consulted?

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Capstone Dataset and Baseline case and show the deciding evidence.

Capstone has60 independent leaves, several photos per leaf, healthy/spotted labels.

Plan grouping and **baseline**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 26

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Your capstone dataset includes duplicate photographs in both **training** and validation. What should be repaired before measuring the **baseline**?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 34 Test and reflect

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Capstone Dataset and Baseline adds to Responsible Plant Classifier and keep the evidence.

Publish the frozen Plant Classifier design-and-data bundle with version IDs and approval record.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and **baseline** procedure. When may final labels be consulted?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 State a question

> State a question. Apply the supplied Capstone Dataset and Baseline facts and keep your reasoning.

Write a bounded question and a non-use statement for the supplied task.

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Define fields

> Define fields. Apply the supplied Capstone Dataset and Baseline facts and keep your reasoning.

Explain why record ID, group ID and measured width have different roles.

My reasoning and evidence

WORKBOOK · 14 OF 26

W3 Set criteria

> Set criteria. Apply the supplied Capstone Dataset and Baseline facts and keep your reasoning.

Name three acceptance criteria that do not require the candidate to win.

My reasoning and evidence

WORKBOOK · 15 OF 26

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Capstone Dataset and Baseline facts and keep your reasoning.

A proposed 80-row dataset has 40 related pairs, and the team splits individual rows randomly then uses pair ID as a feature. Revise its plan.

My reasoning and evidence

WORKBOOK · 16 OF 26

W5 Prepare a fallback

Prepare a fallback. Apply the supplied Capstone Dataset and Baseline facts and keep your reasoning.

What should the team do if it cannot obtain a valid untouched final test?

My reasoning and evidence

WORKBOOK · 17 OF 26

Capstone models and baselines — W1 Build the baseline

Use the supplied Capstone Dataset and Baseline task to record your reasoning, evidence and next step.

State the worked **training**-majority rule and validation accuracy.

> **My reasoning and evidence**

WORKBOOK · 18 OF 26

Capstone models and baselines — W2 Evaluate the candidate

> Use the supplied Capstone Dataset and Baseline task to record your reasoning, evidence and next step.

Compute the worked candidate accuracy, precision and recall.

> **My reasoning and evidence**

WORKBOOK · 19 OF 26

Capstone models and baselines — W3 Explain the tradeoff

> Use the supplied Capstone Dataset and Baseline task to record your reasoning, evidence and next step.

Compare errors rather than only headline accuracy.

My reasoning and evidence

WORKBOOK · 20 OF 26

Capstone models and baselines — W4 Apply independently

> Use the supplied Capstone Dataset and Baseline task to record your reasoning, evidence and next step.

Training 18 clear/6 blocked; validation **baseline**[[9, 0],[3, 0]], candidate[[8, 1],[1, 2]]. Calculate accuracy for each and candidate precision/recall.

My reasoning and evidence

WORKBOOK · 21 OF 26

Capstone models and baselines — W5 Freeze evidence

> Use the supplied Capstone Dataset and Baseline task to record your reasoning, evidence and next step.

List what must be saved before the teacher releases final-test labels.

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

Pictorial case practice: Capstone Dataset and Baseline

> Plan the plant capstone around independent groups and a baseline before collecting more

> rows. Explain the labelled step before checking the changed condition.

Plan the plant capstone around independent groups and a **baseline** before collecting more rows.

Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H. Target is visible-spot status, with review for hidden leaves.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 24 OF 26

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current **milestone**: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plan the plant capstone around independent groups and a **baseline** before collecting more rows.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Fit majority label only from development **training**; record metrics and error types.

1. Define target

2. Split by groups

3. Plan **baseline**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Plan **baseline** Fit majority label only from development **training**; record metrics and error types. Read the picture in order.

Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A derived crop of plantG is added to **training**. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The final group separation is violated.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The final group separation is

> violated. The crop retains the same source plant identity even with a new file name.

Repair a missing building block

Review Model Cards and Human Review (ai-4/33). Goal: Write a model card whose claims match **evidence**. Short practice: Write a bounded replacement for the publicity sentence.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Subgroup Performance (ai-4/31). Goal: Calculate subgroup results with sample sizes. Short practice: Why should the report retain subgroup rates alongside the total?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What comparison is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Capstone Dataset and Baseline picture from its labels, then record a relevant limit.

01. Capstone Dataset and Baseline: the case

AI ACADEMY / AI-4/34 · 01 · LOOK AT THE FACTS

Capstone Dataset and Baseline: the case

The situation

Plan the plant capstone around independent groups and a baseline before collecting more rows.

Supplied evidence

Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H. Target is visible-spot status, with review for hidden leaves.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain capstone dataset and baseline using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

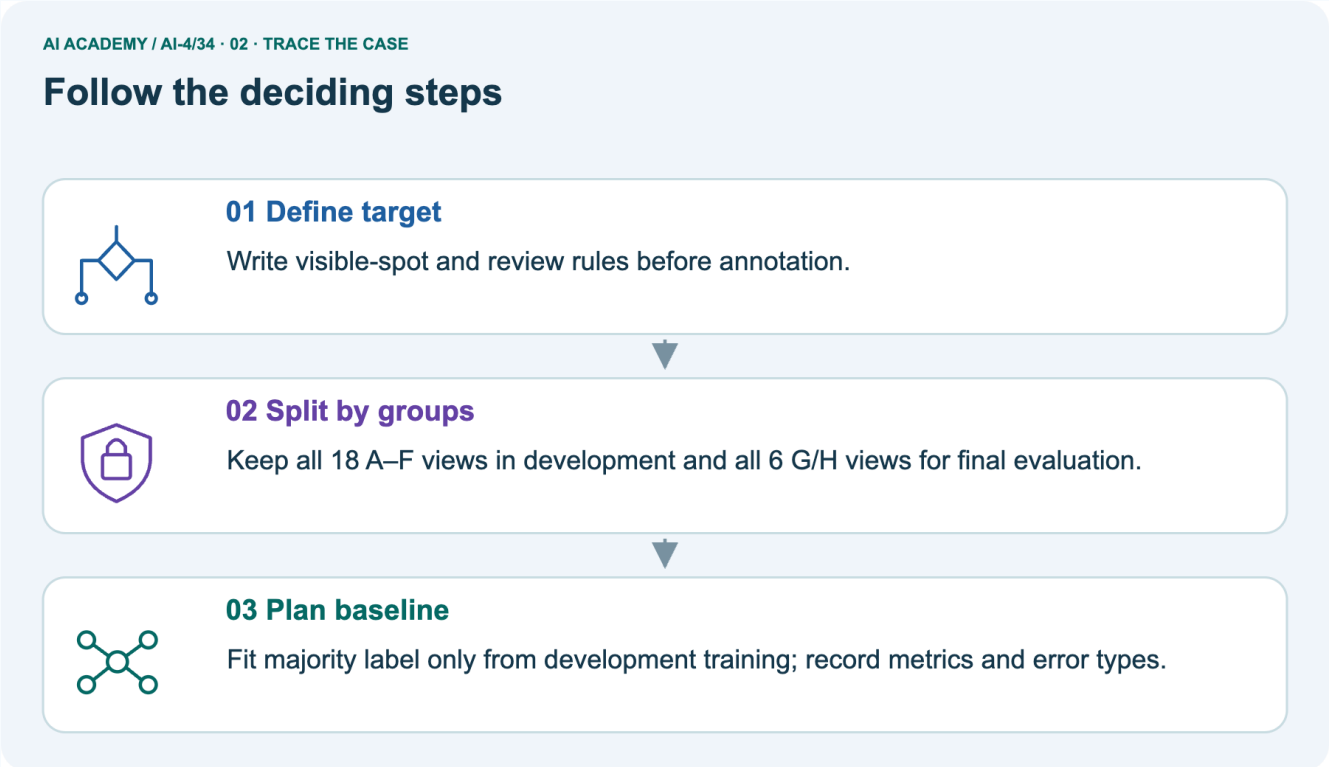
Plan the plant capstone around independent groups and a **baseline** before collecting more rows. Supplied facts: Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H.

Target is visible-spot status, with review for hidden leaves. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Define target: Write visible-spot and review rules before annotation. → Split by groups: Keep all 18 A–F views in development and all 6 G/H views for final evaluation. → Plan **baseline**: Fit majority label only from development **training**; record metrics and error types.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Four-step worked evidence for Week 34: Capstone Dataset and Baseline

AI ACADEMY / AI-4/34 · 03 · INSPECT THE LESSON MODEL

Four-step worked evidence for Week 34: Capstone Dataset and Baseline

Step	Evidence and reasoning
1	Freeze dataset card v3 and grouped split s2 with no plant identity crossing sets.
2	Test set remains sealed; validation supports model choices.
3	Validation has 12 healthy and 8 spotted leaves. The always-healthy baseline scores 12/20=60% accuracy and 0/8=0% spotted recall. The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.
4	Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Follow the labelled picture with the matching explanation. Open the vector diagram

Four-step worked **evidence** for Week 34: Capstone Dataset and **Baseline**. 1 | **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets.. 2 | Test set remains sealed; validation supports model choices.. 3 | Validation has 12 healthy and 8 spotted leaves.

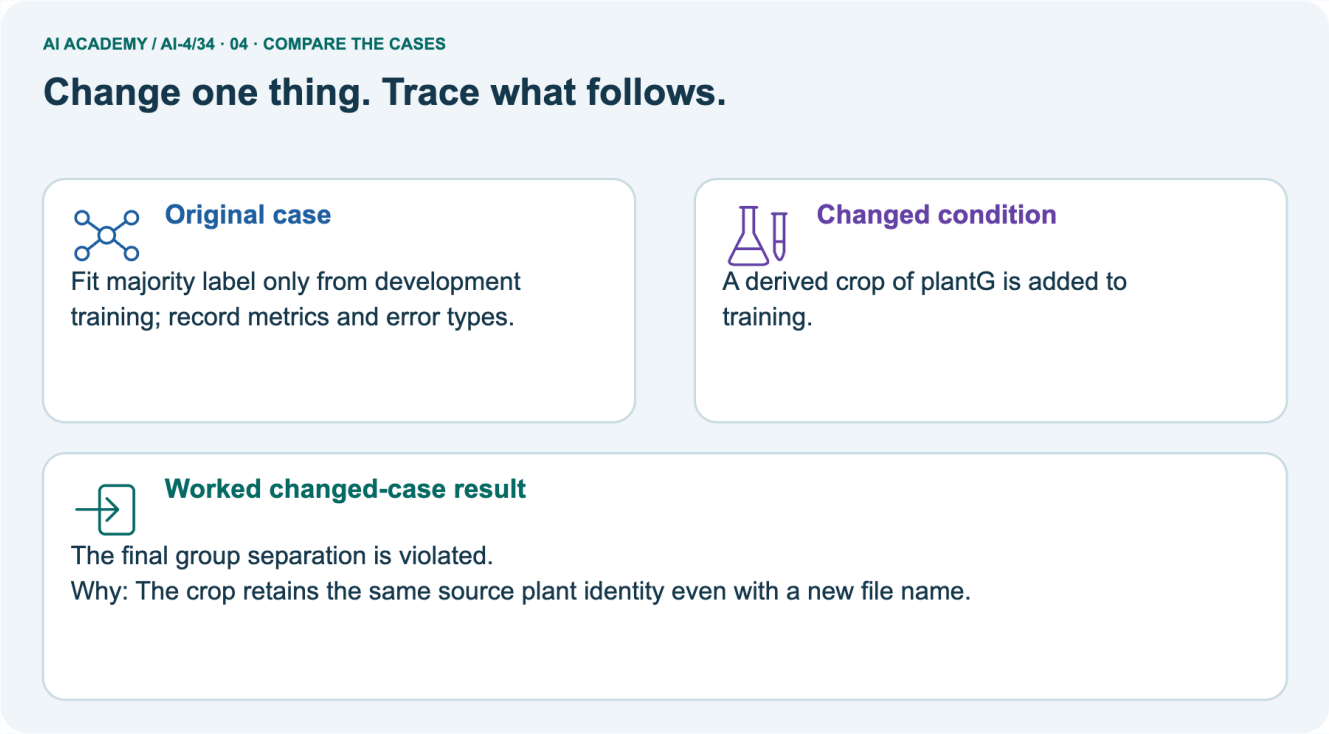
The always-healthy **baseline** scores 12/20=**60%** accuracy and 0/8=0% spotted recall.

The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.. 4 | Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

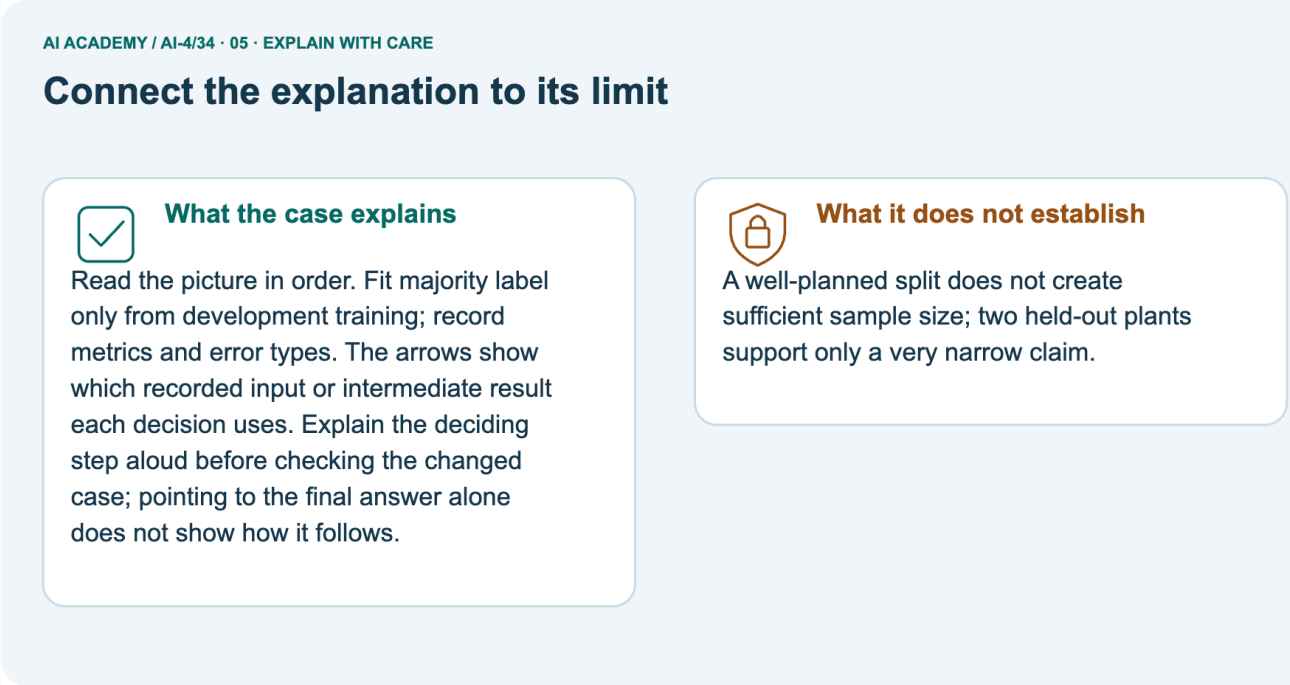
Original case: Fit majority label only from development **training**; record metrics and error types. Changed condition: A derived crop of plantG is added to **training**. Expected result: The final group separation is violated.

Reason: The crop retains the same source plant identity even with a new file name.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

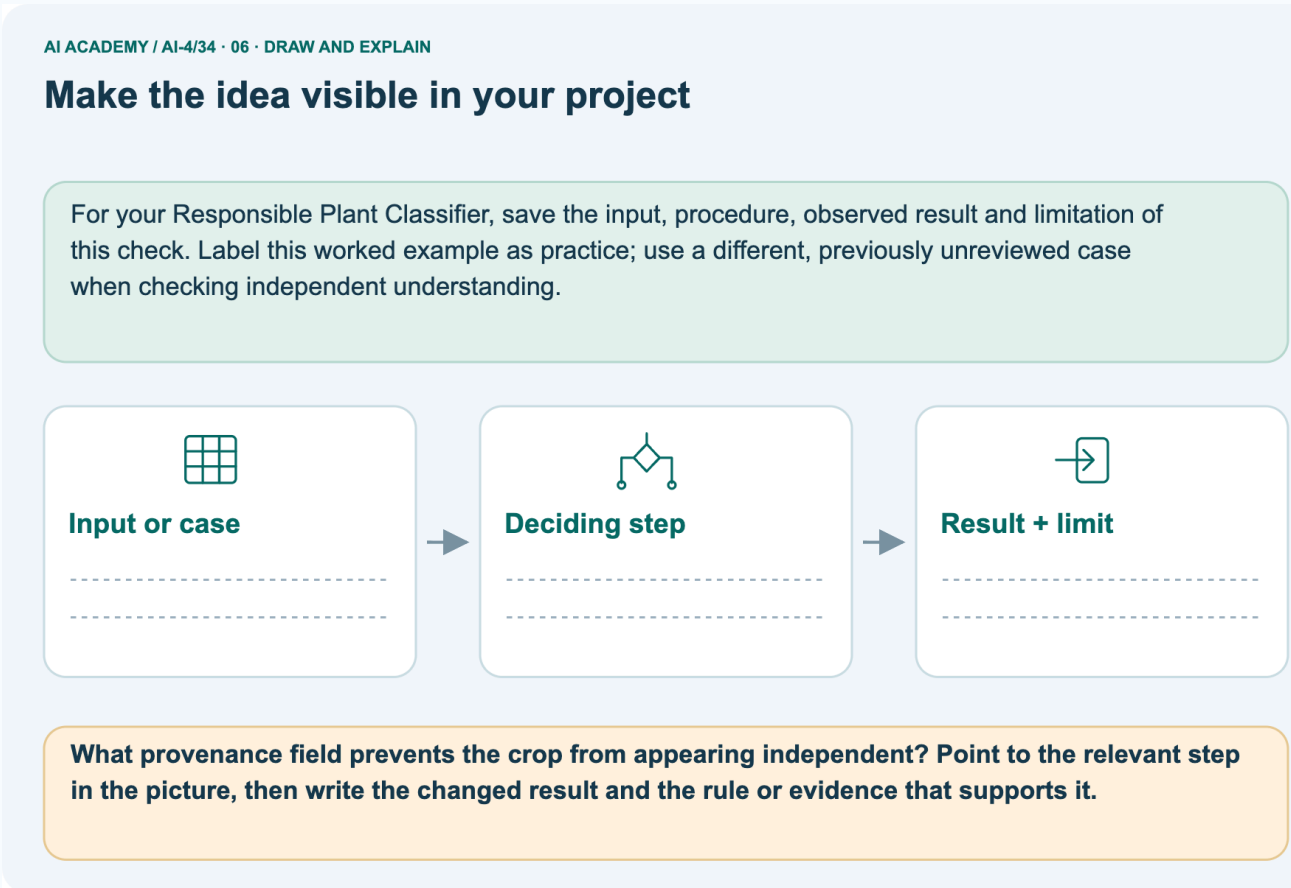
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A well-planned split does not create sufficient sample size; two held-out plants support only a very narrow claim.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.